



Fiji Macro Cheat Sheet



Print Statements

```
print();  
print(myResult);  
print(nSlices);  
print("\\Clear");
```

Comments

```
//Single Line  
/*  
Multi Line  
*/
```

Arrays

```
myArray = newArray(0);  
longArray = Array.concat(arr1, arr2)  
Array.print(arr1)
```

Directory Functions

```
path = getDirectory("Choose Your Directory");  
filelist = getFileList(path);  
resultsDir = path+"_Results";  
File.makeDirectory(resultsDir);
```

Conditional statements

```
if (condition = true) { // Do code  
for (i=0; i<10; i++) { // Do code  
while (condition) { // Do code
```

ROI Manager

```
roiManager("add");  
roiManager("delete");  
roiManager("select", index);  
roiManager("measure");  
roiManager("rename", name);  
roiManager("count");  
roiManager("reset");  
roiManager("deselect");  
roiManager("save", path);  
roiManager("show all");  
roiManager("show none");
```

Common Functions

```
run("Median...", "radius=2");  
run("Z Project...", "projection=[Max Intensity]");  
run("Duplicate, ...");  
run("Duplicate...", "title="+windowtitle+"_copy");  
run("Split Channels");  
run("Measure");  
myAreaResult = getResult("Area", 0);  
setAutoThreshold("Huang no-reset");  
run("Convert to Mask");
```

Resetting

```
close();  
run("Clear Results");  
roiManager("reset");  
while(nImages>0){  
    close();  
}  
print("\\Clear");
```

Current State

```
getVersion();  
nImages;  
getTime();  
getDateAndTime(year, month, dayOfWeek, dayOfMonth,  
hour, minute, second, msec);  
slectionType();  
nResults();  
getResult("column", index);
```



Fiji Macro Cheat Sheet



Operator	Description
++	pre or post increment
--	pre or post decrement
-	minus
!	boolean complement
~	bitwise complement
*	multiplication
/	division
%	remainder
&	bitwise AND
	bitwise OR
^	bitwise XOR
<<, >>	left shift, right shift
+	addition or string concatenation
-	subtraction
<, <=	less than, less than or equal
>, >=	greater than, greater than or equal
==, !=	equal, not equal
&&	boolean AND
	boolean OR
=	assignment
+=, -=, *=, /=	assignment with operation

Stack Control

```
Stack.setDisplayMode("color");
Stack.setDisplayMode("composite");
Stack.setDisplayMode("grayscale");

Stack.setChannel(channel);
Stack.setFrame(Frame);
```

Dialog Control

```
Dialog.create("Title");

Dialog.addMessage("string");
Dialog.addString(label, default);
Dialog.addNumber(label, default);
Dialog.addChoice(label, items);
Dialog.addCheckbox(label, items);

Dialog.show();

Dialog.getString();
Dialog.getNumber();
Dialog.getChoice();
Dialog.getCheckbox();
```

Image Information

```
windowTitle = getTitle();
selectWindow(thisWindow);
getDimensions(width, height, channels, slices, frames);
nSlices;
getPixelSize(unit, pixelWidth, pixelHeight);
getVoxelSize(width, height, depth, unit);
getResult("column", row);
```

Saving & Reporting

```
saveAs("Tiff", resultsDir+"myfilename");
summaryFile = File.open(resultsDir+"Results.csv");
print(summaryFile, "Image Name, ... Results");
```